WAIRAKA MODERN SS-JINJA

S3 END OF TERM 1, 2026

PHYSICS 1

TIME 2hours 15 minutes

INSTRUCTIONS

- This paper consists of ***four items***
- All items are compulsory
- use calculators and diagrams where necessary and clearly show all your workings.

Attempt all items

ITEM ONE

In a certain country, one television (TV) reporter was reporting live near the ocean about the rise and fall in water levels and how it has affected the neighboring communities. Viewers in another country were watching live night broad cast of news bulletin during that time. The viewers wondered how it could be day and night at the same time. Another reporter in Australia was reporting about the dry spell that has caused wild fires while the one in Europe reported about ice and snow that has hindered transportation of people and their goods. The viewers also notice upon looking at the sky that the moon kept on changing shapes within days These puzzled viewers the more.

Task: As a physics learner;

Help the viewers

(a) Understand the possibility of being day in one country and night in another country.

(b) Explain the cause of fall and rise in water levels and how it has affected communities around.

(c) Understand the cause the dry spell in Australia, when there is ice and snow in Europe.

(d) Explain why the moon appeared to be changing its phase

ITEM TWO

A certain driver set-off from his work place with a uniform speed of 20 kmh^{-1} for 1.0 hours. As he approached the traffic light, he decelerated to 15kmh^{-1} for 0.5 hours after which he accelerated to 30kmh^{-1} for 1.0 hours. To his surprise, the car started to decelerate uniformly to rest in 0.5 hours due to lack of fuel and heavily loaded mini-bus by passed him with lots of luggage on top of its roof rack which greatly affects its stability. He attributed this mess of abrupt stopping to the petrol station attendant who might have cheated him yet he paid him money for full tank re-filling. However, when the mechanic came to inspect the car, it was found out that the tyres were hot and almost at the verge of bursting if the journey was to be continued.

Hint:

- Capacity of the car's fuel tank was 20 litres.
- Car consumed $\frac{1}{4}$ liters of fuel per km.

Task: As a learner of Physics;

(a) i) Sketch the velocity-time motion graph for the driver
ii) How long had the Driver travelled?
(b) Prove whether the driver was cheated by the fuel attendant.
(c) Explain how the stability of the by-passing vehicle can be improved.
(d) Explain using Kinetic theory, why the car tyre would burst if the journey was continued.

ITEM THREE

During a study tour, a student who weighs 60kg climbs up a hill of vertical height 100m and finds the rest of the people at the top of the hill. He appears very tired and sweaty. He therefore removes the sweater. The group leader appears not to understand the challenges of this student and wants the group immediately resume the journey. He labels the student as indiscipline.

On his way back home, he boarded a car which started from rest and accelerated uniformly to a velocity of 15 ms^{-2}

Hint: *acceleration due to gravity* $g=10ms^{-2}$

TASK:

Assist the group leader to

a) i) Appreciate the reasons behind the actions of the student.
ii) Work done by the student

b) determine the distance travelled by the car

ITEM FOUR

A group of tourists set off from the bottom of a mountain to climb up to the peak of the mountain. However, on the peak of the mountain, they all complained of difficulty in breathing and eventually experienced nose bleeding. All of the sudden everyone picked up a heavy jacket and

put it on. On using a mercury barometer, a barometric height of 71cm was measured. One of the tourists was told its possible to cook around the mountain but got stuck where exactly.

Hint: Density of mercury 13600 kg m^{-3}

Task

Using your knowledge of physics, Explain why they;

a) (i) Experienced difficulty in breathing and nose bleeding
   (ii) Had to put on heavily jackets.
b) What is the pressure at the peak of the mountain
c) On which side of the mountain is cooking food faster and why?

NEVER GIVE UP

THE END